



COMPETENCY STANDARD
FOR
PLUMBING AND PIPE FITTING
(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Plumbing and Pipe Fitting of is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated to use in training under the Skills for Industry Competitiveness and Innovation Program (SICIP) to meet the industry skills requirements. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity.
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (30 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-CON-PLU-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-CON-PLU-02-G	Carry out workplace interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-CON-PLU-03-G	Operate in a team environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
Total Hour			30

Sector Specific Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-CON-PLU-1-S	Apply green practices in Construction Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-CON-PLU-2-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	20
SICIP-CON-PLU-3-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate construction work measurements 3. Execute simple construction work calculations 4. Clean and maintain measuring instruments	10
Total Hour			40

Occupation Specific Competencies (290 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-PLU-01-O	Perform Pipe Threading Operation	1. Gather and check tools, equipment and materials 2. Carry out pipe cutting operation 3. Carry out thread cutting operation 4. Assemble pipe run 5. Clean/maintain the work area	40
SICIP-CON-PLU-02-O	Perform Access Cutting and Encroachment Works	1. Interpret drawings and plumbing plans 2. Inspect encroachment work area 3. Gather tools, equipment and materials 4. Cut and make access through walls and floors 5. Clean/maintain the work area	40
SICIP-CON-PLU-03-O	Carry Out Water Supply Line Installation Using G.I., PPR and HDPE Pipes	1. Gather and inspect tools, equipment and materials 2. Perform G.I./PPR/HDPE pipe cutting operation 3. Perform Pipe threading operation 4. Assemble pipe runs 5. Clean/maintain the work area	48
SICIP-CON-PLU-04-O	Carry Out Water Supply Line Installation Using uPVC /cPVC Pipes	1. Gather and inspect tools, equipment and materials 2. Perform uPVC /cPVC pipe cutting operation 3. Perform uPVC /cPVC pipe run assembly 4. Clean/maintain the work area	48
SICIP-CON-PLU-05-O	Carry Out Sanitary Pipeline Installation	1. Plan out sanitary pipes installation	36

Code	Unit of Competency	Elements of Competency	Duration (hours)
		2. Gather tools, equipment and materials 3. Carry out trenching and bedding works 4. Install sanitary pipes 5. Finish final pipe run 6. Clean/maintain the work area	
SICIP-CON-PLU-06-O	Carry Out Plumbing Fixtures Installation and Testing	1. Prepare for plumbing fixture installation 2. Gather tools, equipment and materials 3. Install a western or asian commode 4. Install other plumbing fixtures 5. Perform pressure testing 6. Clean/maintain the work area	52
SICIP-CON-PLU-07-O	Carry out Troubleshooting in Plumbing System	1. Identify common plumbing system problems 2. Prepare for troubleshooting work 3. Troubleshoot plumbing system 4. Clean/maintain the work area	26
Total Hours			0290

COMPETENCY STANDARD: PLUMBING AND PIPE FITTING

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-PLU-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 <u>Emergency response plans and procedures</u> are responded to. 1.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron

	2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual 1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-PLU-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
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1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.1 Notes 1.2 Arranging a meeting 1.3 Agenda 1.4 Simple reports such as progress and incident reports 1.5 Job sheets 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 Standards 1.10 OHS information 1.11 Signs
3. Appropriate sources	1.1 Human Resources (HR) Department 1.2 Managers 1.3 Supervisors 1.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	1.9 Commitment to occupational health and safety 1.10 Promptness in carrying out activities 1.11 Sincere and honest to duties 1.12 Environmental concerns 1.13 Eagerness to learn 1.14 Tidiness and timeliness 1.15 Respect for rights of peers and seniors in the workplace 1.16 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities.
4.8	Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-PLU-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	1.1. Identifying own role and responsibilities within team 1.2. Communicating and co-operating with team members 1.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	1.4. Commitment to occupational health and safety 1.5. Promptness in carrying out activities 1.6. Sincere and honest to duties 1.7. Eagerness to learn 1.8. Tidiness and timeliness 1.9. Environmental concerns 1.10. Respect for rights of peers and seniors at workplace 1.11. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.12. Workplace (simulated or actual) 1.13. Workplace documents, Projector 1.14. Learning manual 1.15. Tools, equipment and facilities appropriate to the process or activities. 1.16. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN CONSTRUCTION SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-PLU-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in construction sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Green concepts in the construction sector</u> are understood. 1.2 <u>Sustainable building materials</u> are recognized. 1.3 Energy-efficient designs are interpreted. 1.4 <u>Water conservation practices</u> are understood. 1.5 Construction waste management is recognized. 1.6 <u>Building codes and regulations</u> are interpreted. 1.7 <u>Green building certifications</u> are understood.
2. Minimize resources use in the workplace	2.1 Resource consumption in the construction sector is regularly monitored. 2.2 <u>Energy-efficient equipment and technologies</u> are utilized. 2.3 Material selection is optimized. 2.4 Water usage is managed by using water-saving systems. 2.5 Optimized building designs are implemented. 2.6 Sustainable practices are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in the construction sector. 3.2 Waste segregation is implemented on-site. 3.3 Recycling programs are recognized. 3.4 On-site storage and disposal methods are described. 3.5 Construction processes are followed ensuring waste management practices.

Range of Variables

Variable	Range
	May include but not limited to:
1. Green concepts in the construction sector	1.1 Sustainable Building Materials 1.2 Energy-Efficient Design 1.3 Water Conservation Measures 1.4 Waste Management and Recycling 1.5 Site Environmental Protection

	1.6 Renewable Energy Integration
2. Sustainable building materials	2.1 Recycled and Reclaimed Materials 2.2 Renewable Natural Materials 2.3 Low-Emission and Non-Toxic Materials 2.4 High-Performance Insulation Materials 2.5 Durable and Long-Lasting Materials 2.6 Innovative Eco-Friendly Materials
3. Water conservation practices	3.1 Rainwater Harvesting Systems 3.2 Greywater Recycling 3.3 Low-Flow Fixtures and Fittings 3.4 Efficient Site Water Management 3.5 Landscaping with Drought-Resistant Plants 3.6 Water Leak Detection and Repair Systems 3.7 Use of Water-Efficient Appliances and Equipment
4. Building codes and regulations	4.1 Bangladesh National Building Code (BNBC) 4.2 Environmental Conservation Rules (1997) 4.3 Water Supply and Sewerage Authority (WASA) Regulations 4.4 Fire Service and Civil Defense Regulations 4.5 Bangladesh Labour Act (2006) and Construction Safety 4.6 Local Government (City Corporation and Municipal) Building Bye-laws 4.7 Energy Efficiency and Green Building Guidelines
5. Green building certifications	5.1 LEED (Leadership in Energy and Environmental Design) 5.2 BREEAM (Building Research Establishment Environmental Assessment Method) 5.3 WELL Building Standard 5.4 EDGE (Excellence in Design for Greater Efficiencies) 5.5 Green Globe Certification 5.6 Green Rating for Integrated Habitat Assessment (GRIHA)
6. Energy-efficient equipment and technologies	6.1 High-Efficiency HVAC Systems 6.2 LED and Smart Lighting Systems 6.3 Energy-Efficient Construction Machinery 6.4 Building Automation and Control Systems (BACS) 6.5 Solar-Powered Equipment and Technologies 6.6 Insulation and Building Envelope Technologies
7. Types of waste	7.1 Concrete and Masonry Waste 7.2 Wood Waste 7.3 Metal Waste 7.4 Plastic and Packaging Waste 7.5 Hazardous Waste 7.6 Soil and Excavation Waste 7.7 Glass and Ceramics Waste

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental principles of green construction. 1.2 Benefits, environmental footprint, and lifecycle considerations 1.3 Overview of sustainable building materials 1.4 Relevant building codes and environmental regulations 1.5 Common types of construction waste 1.6 Waste segregation principles and methods 1.7 Green building certifications 1.8 Principles of integrated design and project management 1.9 Sustainable construction techniques
2. Underpinning Skills	2.1 Identifying and selecting sustainable building materials 2.2 Interpreting and applying energy-efficient design principle 2.3 Establishing and managing on-site waste storage systems 2.4 Segregating construction waste on-site into recyclable, reusable, and hazardous categories 2.5 Coordinating with recycling facilities and waste disposal contractors 2.6 Following building codes, environmental regulations, and green certification 2.7 Documenting waste generation and disposal activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace
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	1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-CON-PLU-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools for usability, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after

	use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage. 4.5. Defective hand and power tools are inspected and repaired or replaced. 4.6. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	Hammer 1.1. Bench vice 1.2. Files 1.3. Punches 1.4. Chisels 1.5. Wrenches 1.6. Pliers 1.7. Scriber 1.8. Scraper 1.9. Screwdrivers 1.10. Dividers 1.11. Surface plate 1.12. Gauge 1.13. Tap sets 1.14. Die sets 1.15. Hacksaw 1.16. Socket spanners 1.17. Spanners 1.18. Vice grip 1.19. Wire cutters 1.20. Drills 1.21. Drill bits 1.22. Grinder 1.23. Clamps 1.24. Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine

	2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Testers 3.3 Megger 3.4 Measuring tape 3.5 Hose level 3.6 Water level 3.7 Calipers 3.8 Steel rule 3.9 Meter rule 3.10 Spirit level 3.11 Protractor 3.12 Tri-square 3.13 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on types of hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-PLU-03-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry-out measurements and calculations. It specifically includes the task of checking usability of measuring devices, carrying out accurate construction work measurements, executing simple construction work calculations and cleaning and maintaining measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate <u>measuring device</u> is selected for the job. 1.2 Applications of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate construction work measurements	2.1 Measurements are obtained using appropriate measuring device. 2.2 <u>Systems of measurements</u> are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple construction work calculations	3.1 Simple calculations involving <u>four basic mathematical operations</u> are executed. 3.2 Other operations are used to complete tasks in construction works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign matters are removed from measuring instrument 4.2 Condition of instrument is checked 4.3 Appropriate lubricant is applied after use and prior to storage 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance to workplace procedure

Range of Variables

Variable	Range May include but not limited to:
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1. Measuring device	1.1 Slide calipers 1.2 Steel tape measure 1.3 Steel rule 1.4 Square 1.5 Tri-square 1.6 Feeler gauges 1.7 Water level 1.8 Hose level 1.9 Thermometers 1.10 Protractor
2. Systems of measurements	2.1 ISO standard 2.2 English system 2.3 Metric system
3. Four basic mathematical operations	3.1 Addition 3.2 Subtraction 3.3 Multiplication 3.4 Division

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the

	workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Projector 4.4 Learning manual 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: PERFORM PIPE THREADING OPERATION	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-PLU-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Perform Pipe Threading Operation. It specifically includes the task of gathering and checking tools, equipment and materials, carrying out pipe cutting operation, carrying out thread cutting operation, assembling pipe run and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather and check tools, equipment and materials	1.1 Required pipe size and dimensions are identified in accordance to plumbing plan/design. 1.2 Pipes are selected and gathered in accordance to specification. 1.3 <u>PPEs</u> are checked and used appropriately. 1.4 <u>Tools and equipment and materials</u> are selected, gathered and checked for usability.
2. Carry out pipe cutting operation	2.1 <u>Pipes</u> are measured and marked in accordance to plumbing plan/drawing specification. 2.2 Pipe is clamped using appropriate clamping device. 2.3 Pipe run length is measured and cut using appropriate cutting tool. 2.4 Pipe length is measured and cut in accordance with plumbing plan/drawing and the type of attachment <u>fittings</u> . 2.5 Pipe run length is measured within the specified tolerance.
3. Carry out thread cutting operation	3.1 Pipes are clamped and fixed using pipe vice. 3.2 Newly cut pipe is de burred/removed of burr using appropriate <u>de burring tool</u> . 3.3 Diestocks are adjusted and initiated into the pipe end squarely. 3.4 Thread cutting is carried out on pipes in accordance to workplace procedure. 3.5 <u>Coolant</u> is used during thread cutting operation. 3.6 Threads are checked to conform with the specified form and measurement in accordance with plan/drawing.
4. Assemble pipe run	4.1 Pipes are laid down in accordance with the planned/designed pipe run. 4.2 Pipes and pipe fittings are assembled and fixed in

	accordance with planned/designed pipe run. 4.3 Pipes, fittings and pipe runs are checked for damage/quality of work. 4.4 Assembled pipe runs are measured/checked within tolerable dimensions/length in accordance with design/specification.
5. Clean/maintain the work area	5.1 Plumbing tools/equipment are cleaned and checked for operability. 5.2 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. PPE	1.1 Safety helmet 1.2 Safety shoes 1.3 Hand gloves 1.4 Safety glass 1.5 Apron 1.6 Appropriate clothing
2. Tools, equipment and materials	2.1 Tools: 2.1.1 Measuring tape (5 mtrs.) 2.1.2 Steel rule 2.1.3 Caliper 2.1.4 Adjustable wrench 2.1.5 Box wrench set 2.1.6 Pipe wrench 2.1.7 Pliers, mechanical 2.1.8 Die stock 2.1.9 Pipe cutter 2.1.10 Hand hacksaw 2.1.11 Screw driver: Star, flat, positive 2.1.12 Spirit level 2.1.13 Pipe reamer 2.1.14 Files: flat, half round, round 2.2 Equipment 2.2.1 Pipe threading machine 2.2.2 Power hacksaw 2.2.3 Drill press 2.2.4 Grinder, angle 2.2.5 Welding machine 2.3 Materials 2.3.1 Coolant, water based 2.3.2 Soft stone (Marker) 2.3.3 Marking pen 2.3.4 Pencil 2.3.5 Solvent

	2.3.6 Thread tape (Teflon) 2.3.7 Welding rods 2.3.8 Masking tape
3. Pipes	3.1 G.I. pipe 3.2 PVC pipe 3.3 Stainless steel pipe 3.4 B.I. pipe 3.5 Cast iron pipe
4. Fittings	4.1 Elbow, 90° 4.2 Elbow, 45° 4.3 Tee 4.4 Coupling 4.5 Pipe nipple 4.6 Pipe plug
5. De burring tool	5.1 Flat file 5.2 Round file 5.3 Half round file 5.4 Pipe reamer 5.5 Sand paper
6. Coolant	6.1 Water 6.2 Oil 6.3 Air 6.4 Synthetic 6.5 Nitrogen

Curricular Content Guide

1. Underpinning Knowledge	1.1 Interpreting plumbing drawing 1.2 Types of measuring and checking tools 1.3 Measuring and marking techniques on pipes 1.4 Methods of clamping and fixing steel pipes using pipe vice 1.5 Types of pipe cutting tool 1.6 Procedure of cutting pipes using appropriate cutting tool 1.7 De burring tools and their application 1.8 Types of pipe threads 1.9 Purpose of cutting coolants 1.10 Types and function of thread cutting machines 1.11 Procedure of cutting threads on pipes 1.12 Types of pipe fittings and their application 1.13 Methods and techniques of fixing pipes and pipe fittings 1.14 Pipe run calculation method
2. Underpinning Skills	2.1 Measuring and marking pipes 2.2 Clamping and fixing steel pipes using pipe vice 2.3 Cutting pipes using appropriate cutting tool 2.4 De burring newly cut pipe from burr/sharp edges using

	appropriate de burring tool 2.5 Cutting threads on pipes in accordance with workplace procedure 2.6 Fixing pipes and pipe fittings in accordance with planned/designed pipe run. 2.7 Measuring/checking assembled pipe runs within tolerable dimensions/length in accordance with design/specification] 2.8 Cleaning workplace and disposing waste materials in accordance with workplace requirement
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Materials 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Measured and marked pipes in accordance to plumbing plan/drawing specification. 1.2 Clamped and fixed Steel pipes using pipe vice. 1.3 Cut pipes using appropriate cutting tool. 1.4 De burred newly cut pipe from burr/sharp edges using appropriate de burring tool. 1.5 Cut threads on pipes in accordance with workplace procedure. 1.6 Fixed pipes and pipe fittings in accordance with planned/designed pipe run. 1.7 Measured/checked assembled pipe runs within tolerable dimensions/length in accordance with design/specification. 1.8 Cleaned workplace and disposed waste materials in accordance with workplace requirement.
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM ACCESS CUTTING AND ENCROACHMENT WORKS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-PLU-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Perform Access Cutting and Encroachment Works. It specifically includes the task of interpreting drawings and plumbing plans, inspecting encroachment work area, gathering tools, equipment and materials, cutting and making access through walls and floors and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret drawings and plumbing plans	1.1 Building <u>drawing/plumbing plan</u> is gathered. 1.2 Building drawing/plumbing plans are interpreted. 1.3 Work area/ <u>line of encroachment</u> is identified.
2. Inspect encroachment work area	2.1 Possible <u>obstructions/limitations</u> along the line of encroachment work are identified. 2.2 Possible solutions/remedies needed for identified obstructions/limitations are planned out. 2.3 Obstructions/limitations are reported with immediate superior whenever necessary. 2.4 <u>Tools, equipment and materials</u> required to carry out encroachment work are identified.
3. Gather tools, equipment and materials	3.1 Tools, equipment and materials are selected, gathered and checked for usability. 3.2 <u>PPEs</u> are checked and used appropriately.
4. Cut and make access through walls and floors	4.1 Appropriate place for the pipe access work is marked in accordance with the plumbing plan/ specification 4.2 Concrete wall and floor are cut using core cutting machine to create pipe access in accordance to plumbing plan/specification. 4.3 Walls and floors are grooved without causing damage to the wall or floor and adjacent installations. 4.4 Correct usage of tools and equipment is observed.
5. Clean/maintain the work area	5.1 Used tools and equipment are cleaned and maintained. 5.2 Work area is cleaned. 5.3 Excess and waste materials are disposed in accordance to workplace requirement.

Range of Variables

Variable	Range (Includes but not limited to):
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1. Drawing/Plumbing plan	1.1 Plan for water supply system 1.2 Plan for sewerage system 1.3 Plan for Rainwater harvesting system
2. Line of encroachment	2.1 Floors 2.2 Walls 2.3 Ceilings 2.4 Slabs 2.5 Beams 2.6 Columns
3. Obstructions/limitations	3.1 Built in appliances 3.2 Cabinets 3.3 Concrete column 3.4 Concrete beam 3.5 Existing pipe runs 3.6 Door 3.7 Window
4. Tools, equipment and materials	4.1 Tools; 4.1.1 Measuring tape (10 mtrs.) 4.1.2 Steel rule 4.1.3 Cold chisel 4.1.4 Hammer, claw 4.1.5 Mallet 4.1.6 Cold chisel 4.1.7 Adjustable wrench 4.1.8 Box wrench set 4.1.9 Pipe wrench 4.1.10 Pliers, mechanical 4.1.11 Hand hacksaw 4.1.12 Spirit level 4.1.13 Hand hack saw 4.2 Equipment 4.2.1 Core cutting machine 4.2.2 Jackhammer 4.2.3 Drill press 4.2.4 Grinder, angle 4.2.5 Portable drill 4.2.6 Oxy-acetylene cutting outfit 4.3 Materials 4.3.1 Soft stone (Marker) 4.3.2 Marking pen 4.3.3 Pencil 4.3.4 Welding rods 4.3.5 Cement 4.3.6 Sand 4.3.7 Putty 4.3.8 Rubber cement
5. PPE	5.1 Safety helmet

	5.2 Safety shoes 5.3 Hand gloves 5.4 Safety glass 5.5 Apron 5.6 Appropriate clothing
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Methods and procedure of cutting concrete wall and floor to create pipe access 1.2 Interpreting techniques for building drawing/plumbing plans for water supply, sewerage and rainwater harvesting system 1.3 Common types of obstructions/limitations for encroachment work in plumbing 1.4 Workplace policies on tools borrowing, checking and maintenance 1.5 Proper use of PPEs 1.6 Proper use of Core cutting Machine and safety 1.7 Methods of laying-out for access/encroachment work 1.8 Procedure for Cutting of walls and floors without causing damage to the wall or floor and adjacent installations. 1.9 Workplace procedure in Cleaning used tools & equipment and excess and waste materials disposal
2. Underpinning Skills	2.1 Interpreting building drawing/plumbing plans 2.2 Planning-out possible solutions/remedies needed for identified obstructions/limitations for encroachment work 2.3 Selecting, gathering and checking tools, equipment, PPEs and materials for usability 2.4 Laying-out for access/encroachment work 2.5 Cutting walls and floors without causing damage to the wall or floor and adjacent installations 2.6 Cutting Concrete wall and floor using core cutting machine 2.7 Cleaning used tools & equipment and disposed excess and waste materials are in accordance to workplace requirement
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Tools and equipment 4.3 Materials 4.4 Projector 4.5 Stationary 4.6 Learning manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted building drawing/plumbing plans. 1.2 Planned out possible solutions/remedies needed for identified obstructions/limitations for encroachment work. 1.3 Selected, gathered and checked tools, equipment, PPEs and materials for usability. 1.4 Laid-out for access/encroachment work. 1.5 Cut walls and floors without causing damage to the wall or floor and adjacent installations. 1.6 Cleaned used tools and equipment and disposed excess and waste materials in accordance to workplace requirement.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WATER SUPPLY LINE INSTALLATION USING G.I. PPR AND HDPE PIPES	Nominal Duration: 48 Hrs.	Unit Code: SICIP-CON-PLU-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out water supply line installation using G.I.PPR and HDPE Pipes. It specifically includes the tasks of gathering and inspecting tools, equipment and materials, performing G.I/PPR/HDPE pipe cutting operation, performing pipe threading operation, assembling pipe runs and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather and inspect tools, equipment and materials	1.1 <u>PPE</u> are selected and used. 1.2 <u>Tools, equipment and materials</u> are gathered and checked for usability. 1.3 Pipes are identified and gathered in accordance with plumbing plan specifications.
2. Perform G.I/PPR/HDPE pipe cutting operation	2.1 G.I/PPR/HDPE pipes are measured and marked as per plan/drawing specification. 2.2 G.I/PPR/HDPE pipes are clamped using appropriate <u>clamping device</u> . 2.3 G.I/PPR/HDPE pipes are cut using appropriate cutting tool. 2.4 G.I/PPR/HDPE pipes are cut within the specified dimension and considering specified tolerance.
3. Perform Pipe threading operation	3.1 G.I /PPR/HDPE pipes are clamped and fixed using appropriate clamping device. 3.2 Diestocks are adjusted and initiated into the pipe end squarely. 3.3 Thread cutting is carried out on pipes in accordance to workplace procedure. 3.4 <u>Coolant</u> is used during thread cutting operation. 3.5 Threads are checked to conform with the specified form and measurement in accordance with plan/drawing.
4. Assemble pipe runs	4.1 G.I/PPR/HDPE Pipe runs are positioned on the corresponding areas as per plumbing plan. 4.2 Piping joints and fittings are tightened using appropriate tools and <u>sealant</u> . 4.3 <u>Pipe clamps and fixtures</u> along pipe runs are installed in accordance with plumbing plan. 4.4 Leaks and <u>non-conformance</u> to plumbing design/plans are checked and corrected.

	4.5 Holes and openings are filled with cement plasters and <u>finishing.</u>
5. Clean/maintain the work area	5.1 Used tools and equipment are cleaned and maintained. 5.2 Workplace is cleaned. 5.3 Waste materials are disposed in accordance to workplace requirement.

Range of Variables

Variable	Range (Includes but not limited to):
1. PPE	1.1 Safety helmet 1.2 Safety shoes 1.3 Hand gloves 1.4 Safety glass 1.5 Safety Belt 1.6 Apron 1.7 Appropriate clothing
2. Tools, equipment & materials	2.1 Tools; 2.1.1 Measuring tape (5 mtrs.) 2.1.2 Steel rule 2.1.3 Vernier caliper 2.1.4 Adjustable wrench 2.1.5 Box wrench set 2.1.6 Pipe wrench 2.1.7 Pliers, mechanical 2.1.8 Die stock 2.1.9 Tube cutter 2.1.10 Hand hacksaw 2.1.11 Screw driver: Star, flat, positive 2.1.12 Files 2.1.13 Spirit level 2.1.14 Pipe reamer 2.2 Equipment 2.2.1 Pipe threading machine 2.2.2 Power hacksaw 2.2.3 Drill press 2.2.4 Grinder, angle 2.2.5 Welding machine 2.3 Materials 2.3.1 Coolant, water based 2.3.2 Soft stone (Marker) 2.3.3 Marking pen 2.3.4 Pencil 2.3.5 Plastic based sealant 2.3.6 Rubber based sealant

	2.3.7 Synthetic sealant 2.3.8 Thread tape (Teflon) 2.3.9 Welding rods 2.3.10 Pipes 2.3.11 Types of fittings like Gate valve, Check Valve, Globe Valve, Foot Valve etc
3. Clamping device	3.1 Pipe stand 3.2 Bench vise w/v-blocks 3.3 Vise grip 3.4 Pipe chuck 3.5 Threading machine chucks
4. Coolant	4.1 Water 4.2 Oil 4.3 Air 4.4 Synthetic 4.5 Nitrogen
5. Sealants	5.1 Thread tape (Teflon) 5.2 Cement 5.3 Plastic-based seals 5.4 Rubber-based seals 5.5 Synthetic seals 5.6 Silicone seals
6. Pipe clamps and fixtures	6.1 Steel pipe hangers 6.2 Plastic pipe hangers 6.3 Pipe clamps (plastic, steel, rubber) 6.4 Tapes
7. Non-conformance	7.1 Leak 7.2 Length (Dimension) 7.3 Pipe size 7.4 Pipe offset angles 7.5 Obstacles 7.6 Aesthetics/appearance
8. Finishing	8.1 Caulk 8.2 Cement 8.3 Paint 8.4 Tiles/marble 8.5 Grommet
9. PPE	9.1 Safety helmet 9.2 Safety shoes 9.3 Hand gloves 9.4 Safety glass 9.5 Safety Belt 9.6 Apron 9.7 Appropriate clothing

Curricular Content Guide

1. Underpinning	1.1 Types of pipe threads and their nomenclature
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Knowledge	1.2 Techniques of measuring and marking pipes 1.3 Procedure and technique of cutting G.I/PPR/HDPE Pipes 1.4 G.I. pipe thread cutting procedure and proper use of thread cutting tools 1.5 Method of positioning and lay outing G.I/PPR/HDPE pipe runs 1.6 Techniques and safety precautions of tightening G.I/PPR/HDPE piping joints and fittings using sealants 1.7 Thread sealing techniques 1.8 Types, methods and techniques of filling holes and openings after plumbing works 1.9 Types of pipe clamps and fixtures and means of Installation 1.10 Procedure in checking and correcting leaks 1.11 Methods of Filling oversized holes and openings with cement plasters and other finishing materials
2. Underpinning Skills	2.1 Identifying and gathering pipes in accordance with specified sizes/schedules 2.2 Cutting G.I/PPR/HDPE Pipes using appropriate cutting tool and within the specified dimension considering specified tolerance 2.3 Use of PPR welding Machine 2.4 Carrying out thread cutting on pipes in accordance to workplace procedure 2.5 Positioning G.I/PPR/HDPE Pipe runs at the corresponding areas as per plumbing plan 2.6 Tightening piping joints and fittings using appropriate tools and sealant 2.7 Installed clamps and fixtures along pipe runs in accordance with plumbing plan 2.8 Checking and correcting leaks and non-conformance to plumbing design/plans are 2.9 Filling oversized holes and openings with cement plasters and finishing materials 2.10 Cleaning used tools & equipment and disposing excess and waste materials in accordance to workplace requirement
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE)

	4.3	Tools and equipment
	4.4	Materials
	4.5	Projector
	4.6	Stationery
	4.7	Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified and gathered pipes in accordance with specified sizes/schedules. 1.2 Cut G.I/PPR/HDPE Pipes using appropriate cutting tool and within the specified dimension considering specified tolerance. 1.3 Carried out thread cutting on pipes in accordance to workplace procedure. 1.4 Positioned G.I/PPR/HDPE Pipe runs at the corresponding areas as per plumbing plan. 1.5 Tightened piping joints and fittings using appropriate tools and sealant. 1.6 Installed clamps and fixtures along pipe runs in accordance with plumbing plan. 1.7 Checked and corrected leaks and non-conformance to plumbing design/plans. 1.8 Filled oversized holes and openings with cement plasters and finishing materials. 1.9 Cleaned used tools & equipment and disposed excess and waste materials in accordance to workplace requirement.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WATER SUPPLY LINE INSTALLATION USING UPVC /CPVC PIPES	Nominal Duration: 48 Hrs.	Unit Code: SICIP-CON-PLU-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out water supply line installation using uPVC /cPVC Pipes. It specifically includes the tasks of gathering and inspecting tools, equipment and materials, performing uPVC /cPVC pipe cutting operation, performing uPVC /cPVC pipe run assembly and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather and inspect tools, equipment and materials	1.1 <u>PPE</u> are selected and used. 1.2 <u>Tools, equipment and materials</u> are gathered and checked for usability. 1.3 uPVC/cPVC pipe sizes/schedules are identified and gathered.
2. Perform uPVC /cPVC pipe cutting operation	2.1 uPVC/cPVC C pipes are measured and marked as per plan/drawing specification. 2.2 uPVC/cPVC pipes are clamped using appropriate clamping device. 2.3 uPVC/cPVC pipes are cut using appropriate cutting tool. 2.4 uPVC/cPVC pipes are cut within the specified dimension and considering specified tolerance.
3. Perform uPVC /cPVC pipe run assembly	3.1 uPVC/cPVC pipe runs are positioned on the corresponding location as per plumbing plan. 3.2 uPVC/cPVC piping joints and fittings are assembled using appropriate tools and <u>sealant</u> . 3.3 Appropriate curing time for sealant after assembly of pipe joints and fittings is observed to create strong bond. 3.4 <u>Clamps and fixtures</u> along pipe runs are installed in accordance with plumbing plan. 3.5 Leaks and <u>non-conformance</u> to plumbing design/plans are checked and corrected. 3.6 Holes and openings are filled with cement plasters and <u>finishing</u> .
4. Clean/maintain the work area	4.1 Workplace, tools and equipment are cleaned and checked for normal operation. 4.2 Waste materials are disposed in accordance to workplace requirement

Range of Variables

Variable	Range (Includes but not limited to):
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1. PPE	1.1 Safety helmet 1.2 Safety glass (google) 1.3 Safety shoes 1.4 Hand gloves 1.5 Safety Belt 1.6 Safety glass 1.7 Apron 1.8 Appropriate clothing
2. Tools, equipment & materials	2.1 Tools: 2.1.1 Measuring tape (5 mtrs.) 2.1.2 Steel rule 2.1.3 Vernier caliper 2.1.4 Adjustable wrench 2.1.5 Box wrench set 2.1.6 Pipe wrench 2.1.7 Pliers, mechanical 2.1.8 Tube cutter 2.1.9 Hand hacksaw 2.1.10 Screw driver: Star, flat, positive 2.1.11 Files 2.1.12 Spirit level 2.1.13 Pipe reamer 2.2 Equipment 2.2.1 Power hacksaw 2.2.2 Drill press 2.2.3 Grinder, angle 2.3 Materials 2.3.1 Coolant, water based 2.3.2 Soft stone (Marker) 2.3.3 Marking pen 2.3.4 Pencil 2.3.5 Solvent 2.3.6 Silicon sealant
3. Sealants	3.1 Thread tape (Teflon) 3.2 Solvents 3.3 Cement 3.4 Plastic seals 3.5 Rubberized seals 3.6 Synthetic seals
4. Clamps and fixtures	4.1 Plastic C-clamp 4.2 Metallic C-clamp 4.3 Pipe hanger 4.4 Pipe sledge 4.5 Tapes
5. Non-conformance	5.1 Leaks 5.2 Length

	5.3 Pipe size 5.4 Pipe offset angles 5.5 Obstacles 5.6 Aesthetics/appearance
6. Finishing	6.1 Caulk 6.2 Cement 6.3 Paint 6.4 Tiles/marble 6.5 Grommet

Curricular Content Guide

1. Underpinning Knowledge	1.1 Methods and procedure of cutting concrete wall and floor to create pipe access 1.2 Techniques of measuring and marking pipes 1.3 Pipe and thread cutting procedure and appropriate cutting tools 1.4 Types of pipe threads and their nomenclature 1.5 Method of positioning and lay outting steel pipe runs 1.6 Techniques and safety precautions of tightening steel piping joints and fittings using sealants 1.7 Thread sealing techniques 1.8 Methods and techniques of Positioning and lay outting uPVC/cPVC pipe runs 1.9 Tightening procedures for uPVC/cPVC piping joints and fittings using solvents and curing 1.10 Types, methods and techniques of filling holes and openings after plumbing works
2. Underpinning Skills	2.1 Measuring uPVC/cPVC pipes and marking 2.2 Clamping uPVC/cPVC pipes 2.3 Cutting uPVC/cPVC pipes 2.4 Cutting uPVC/cPVC pipes within the specified dimension and considering specified tolerance 2.5 Positioning uPVC/cPVC pipe runs on the corresponding location as per plumbing plan 2.6 Assembling uPVC/cPVC piping joints and fittings using appropriate tools and sealant 2.7 Observing appropriate curing time for sealant after assembly of pipe joints and fittings to create strong bond 2.8 Installing pipe clamps and fixtures on pipe runs in accordance with plumbing plan 2.9 Checking and correcting leaks and non-conformance to plumbing design/plans 2.10 Filling holes and openings with cement plasters and finishing 2.11 Cleaning used tools & equipment
3. Underpinning	3.1 Commitment to occupational health and safety

Attitudes	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Clamped and cut uPVC/cPVC pipes using appropriate clamping device and cutting tool. 1.2 Observed specified dimension and tolerance when measuring and cutting pipes. 1.3 Positioned uPVC/cPVC pipe runs on the corresponding location as per plumbing plan. 1.4 Assembled uPVC/cPVC piping joints and fittings using appropriate tools and sealant. 1.5 Observed appropriate curing time for sealant after assembly of pipe joints and fittings to create strong bond. 1.6 Installed pipe clamps and fixtures on pipe runs in accordance with plumbing plan. 1.7 Checking and correcting leaks and non-conformance to plumbing design/plans. 1.8 Filled holes and openings with cement plasters and finishing.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT SANITARY PIPELINE INSTALLATION	Nominal Duration: 36 Hrs.	Unit Code: SICIP-CON-PLU-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out sanitary pipeline installation. It specifically includes the tasks of planning out sanitary pipes installation, gathering tools, equipment and materials, carrying out trenching and bedding works, installing sanitary pipes, finishing final pipe run and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan out sanitary pipes installation	1.1 <u>Work activities</u> are confirmed by making site visit/inspection. 1.2 <u>Sanitary pipe</u> runs and elevation are identified in accordance with plumbing plans/specification. 1.3 Pipe line slope/pitch is determined in accordance with plumbing plans/specification and result of site visit.
2. Gather tools, equipment and materials	2.1 Required tools, equipment and materials are identified in accordance with plumbing plan and result of site visit/inspection. 2.2 <u>Sanitary pipe</u> materials, sizes/schedules are identified and gathered. 2.3 <u>PPE</u> are selected and used. 2.4 <u>Tools, equipment</u> and materials are gathered and checked for usability.
3. Carry out trenching and bedding works	3.1 Trench/excavation area/line is laid out in accordance with plumbing plan and result of site visit. 3.2 Size of trench/excavation is made in accordance with workplace and plumbing plan requirement. 3.3 Grade/slope of trench is applied in accordance with workplace and plumbing plan requirement. 3.4 <u>Bedding material</u> is laid in accordance with workplace and plumbing plan requirement.
4. Install sanitary pipes	4.1 Sewer pipes are laid on the trench in accordance with workplace and plumbing plan requirements. 4.2 Bell ended sewer pipes are laid by placing the bell end at the uphill side of the pipe run. 4.3 PVC sewer pipes are installed by using appropriate <u>sealing/ gluing materials.</u> 4.4 Soil and vent pipes are placed in accordance with workplace and plumbing plan requirements. 4.5 Alignment and pipe elevation are checked in accordance to workplace and plumbing plan requirements.
5. Finish final pipe run	5.1 Fittings are installed to complete the final run of the sewer

	pipe installation. 5.2 Final pipe run is checked for leaks and <u>non-conformance</u> to workplace and plumbing line requirements. 5.3 Re-works or revision is made where necessary. 5.4 <u>Covering materials</u> are laid on top of the sewer pipe run in accordance with workplace and plumbing plan requirement.
6. Clean/maintain the work area	6.1 Workplace, tools & equipment are cleaned and checked for normal operation. 6.2 Waste materials are disposed in accordance to workplace requirement.

Range of Variables

Variable	Range (Includes but not limited to):
1. Sanitary Pipe	1.1 Sewer pipe 1.2 Soil pipe 1.3 Vent pipe
2. Work activities	2.1 Area clearing 2.2 Traffic re-routing 2.3 Trenching/excavation 2.4 Tools and equipment gathering 2.5 Materials gathering and stacking 2.6 Concreting works
3. PPE	3.1 Safety helmet 3.2 Safety glass (google) 3.3 Safety shoes 3.4 Rubber/plastic boots 3.5 Hand gloves 3.6 Safety glass 3.7 Appropriate clothing
4. Tools and equipment	4.1 Measuring tape (5 mtrs.) 4.2 Adjustable wrench 4.3 Box wrench set 4.4 Pipe wrench 4.5 Pliers, mechanical 4.6 Hand hacksaw 4.7 Screw driver: Star, flat, positive 4.8 Files 4.9 Spirit level 4.10 Pipe reamer 4.11 Cold chisel 4.12 Hammer 4.13 Jackhammer 4.14 Excavating machine 4.15 Concrete cutter 4.16 Power hacksaw

	4.17 Grinder, angle 4.18 Rammer/compactor
5. Bedding material	5.1 Sand 5.2 Gravel 5.3 Concrete 5.4 Bricks
6. sealing/gluing materials	6.1 Solvents 6.2 Cement 6.3 Plastic sealants 6.4 Rubber based sealants 6.5 Synthetic sealants 6.6 PVC primer 6.7 Silicone sealants
7. Non-conformance	7.1 Leaks 7.2 Pipe size 7.3 Fitting size 7.4 Pipe run slope/grade 7.5 Pipe damage
8. Covering materials	8.1 Sand 8.2 Gravel 8.3 Concrete cement

Curricular Content Guide

1. Underpinning Knowledge	1.1 Piping run calculation procedure 1.2 Procedure of Lay outting for trenching and excavation works 1.3 Methods of determining and applying grade/slope for trenching 1.4 Types of bedding materials and their function 1.5 Methods and Procedure for Laying bedding material 1.6 Procedure of Laying sewer pipes on the trench 1.7 Methods and techniques of PVC Sewer pipes sealing/gluing 1.8 Types of gluing and sealing materials used in sewer piping 1.9 Alignment checking techniques for sewer pipes 1.10 Types of sewer fitting and their uses 1.11 Methods and techniques of installing sewer fittings 1.12 Methods and technique if installing Soil and vent pipes 1.13 Leak testing methods and procedures for sewer pipes and fittings
2. Underpinning Skills	2.1. Determining pipe run elevation and slope/pitch in accordance with plumbing plans/specification and result of site visit 2.2. Lay outting the size of trench/excavation in accordance with workplace and plumbing plan requirement 2.3. Applying grade/slope of trench in accordance with

	workplace and plumbing plan requirement 2.4. Laying bedding material in accordance to workplace and plumbing plan requirement 2.5. Laying sewer pipes on the trench in accordance with workplace and plumbing plan requirements 2.6. Installing PVC Sewer pipes by using appropriate sealing/gluing materials 2.7. Checking alignment and pipe elevation in accordance to workplace and plumbing plan requirement 2.8. Installing fittings to complete the final run of the sewer pipe installation 2.9. Installing fittings to complete the final run of the Soil and vent pipes installation 2.10. Checking final pipe run for leaks and non-conformance to workplace and plumbing plan requirements 2.11. Cleaning work area, tools and equipment and proper waste disposal in accordance with workplace policy
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in workplace 3.8. Communication with peers and seniors in workplace
4. Resource Implications	4.1. Workplace (simulated or actual) 4.2. Materials, tools and equipment needed for the activity 4.3. Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Determined pipe run elevation and slope/pitch in accordance with plumbing plans/specification and result of site visit. 1.2 Laid out the size of trench/excavation in accordance with workplace and plumbing plan requirement. 1.3 Applied grade/slope of trench in accordance with workplace and plumbing plan requirement. 1.4 Laid bedding material in accordance to workplace and plumbing plan requirement. 1.5 Laid sewer pipes on the trench in accordance with workplace and plumbing plan requirements. 1.6 Installed PVC Sewer pipes by using appropriate sealing/glueing materials 1.7 Checked alignment and pipe elevation in accordance to workplace and plumbing plan requirement 1.8 Installed fittings to complete the final run of the sewer
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	pipe installation 1.9 Checked final pipe run for leaks and non-conformance to workplace and plumbing plan requirements 1.10 Laid covering materials on top of the sewer pipe run in accordance with workplace and plumbing plan requirement
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT PLUMBING FIXTURES INSTALLATION AND TESTING	Nominal Duration: 52 Hrs.	Unit Code: SICIP-CON-PLU-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out plumbing fixtures installation and testing. It specifically includes the tasks of preparing for plumbing fixture installation, gathering tools, equipment and materials, installing a western or asian commode, installing other plumbing fixtures, performing pressure testing and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for plumbing fixture installation	1.1 <u>Work activities</u> are confirmed in accordance with plumbing plan and workplace requirements. 1.2 Work site is visited and prepared prior to installation works. 1.3 <u>Plumbing fixture</u> is determined, collected and checked for workability and quality.
2. Gather tools, equipment and materials	2.1 Required <u>tools, equipment</u> and materials are identified in accordance with plumbing plan and result of site visit/inspection. 2.2 Plumbing fixtures to be installed are gathered in the work site. 2.3 <u>PPEs</u> are selected and used. 2.4 Tools, equipment and materials are gathered and checked for usability.
3. Install a western or asian commode	3.1 The installation site is prepared, ensuring a level surface and proper alignment with the drainage outlet or flange. 3.2 The commode is positioned correctly, ensuring alignment with the waste outlet or floor fixtures, and a wax ring is placed for Western commodes. 3.3 The commode is secured using appropriate fasteners, ensuring bolts or fixing mechanisms are tightened without over-tightening. 3.4 <u>Sealing material</u> is applied around the base of the commode according to workplace standards. 3.5 The water supply connections are established securely, and for Western commodes, the water closet tank is installed following manufacturer instructions. 3.6 The installation is inspected for leaks, stability, and proper connection to drainage and water supply systems. 3.7 Any necessary adjustments are made to ensure the commode is correctly aligned and fully operational,

	adhering to workplace and safety requirements.
4. Install other plumbing fixtures	4.1 Plumbing fixtures are installed in accordance with plumbing plan and following manufacturer's instruction/specification. 4.2 Check newly installed plumbing fixtures for leaks and <u>non-conformance</u> with workplace requirements. 4.3 Reworks/adjustment is carried out in accordance with workplace requirements.
5. Perform pressure testing	5.1 <u>Work activities</u> are identified in accordance with plumbing plan and workplace requirements. 5.2 <u>Pressure testing method</u> and maximum test pressure are identified. 5.3 Piping system connections are reviewed and checked for tightness. 5.4 Required <u>tools, equipment and materials for pressure testing</u> are collected. 5.5 Preliminary, intermediate and final test pressures is applied progressively into the system. 5.6 Appropriate <u>Leak testing method</u> is applied to determine presence of leaks. 5.7 Leak testing result is recorded and reported to the immediate superior.
6. Clean/maintain the work area	6.1 Workplace, tools and equipment are cleaned and checked for normal operation. 6.2 Waste materials are disposed in accordance to workplace requirement.

Range of Variables

Variable	Range (Includes but not limited to):
1. Work activities	1.1 Western or Asian commode Shower 1.2 Bath tubs 1.3 Wash basin 1.4 Kitchen sink 1.5 Urinals 1.6 Water closet 1.7 Bidets 1.8 Drinking fountains 1.9 Terminal valves/faucets
2. Plumbing fixture	2.1 Western commode 2.2 Asian commode 2.3 Shower 2.4 Bath tubs 2.5 Wash basin 2.6 Kitchen sink 2.7 Urinals 2.8 Water closet 2.9 Bidet

	2.10 Drinking fountains 2.11 Geyser 2.12 Hot and cold mixer tap 2.13 Terminal valves/faucets
3. Tools and equipment	3.1 Measuring tape (5 mtrs.) 3.2 Adjustable wrench 3.3 Box wrench set 3.4 Pipe wrench 3.5 Pliers, mechanical 3.6 Hand hacksaw 3.7 Screw driver: Star, flat, positive 3.8 Files 3.9 Concrete cutter 3.10 Grinder, angle 3.11 Electric drill
4. PPE	4.1 Hard hat (Skull guard) 4.2 Safety glass (receptacles) 4.3 Hand gloves 4.4 Safety shoes 4.5 Apron 4.6 Working clothes
5. Sealing material	5.1 Concrete/Cement 5.2 Rubber grommet 5.3 Plastic grommet 5.4 White cement 5.5 Wax seal 5.6 Rubber-based seal 5.7 Silicone seal 5.8 Plastic sealants 5.9 Synthetic sealants
6. Non-conformance	6.1 Leaks 6.2 Damaged fixtures 6.3 Broken/damaged fittings 6.4 Wrong Pipe size 6.5 Wrong Fitting size 6.6 Damage flooring/walls
7. Work activities	7.1 Checking tightness of valves/fittings 7.2 Measuring pipe runs 7.3 Identifying piping run which are subject for pressure testing 7.4 Identifying fittings to be isolated 7.5 Isolating system components
8. Pressure testing method	8.1 Pneumatic testing method 8.2 Hydrostatic testing
9. Tools, equipment and materials	9.1 Measuring tape 9.2 Adjustable wrench 9.3 Box wrench set

	9.4 Pipe wrench 9.5 Pliers, mechanical 9.6 Screw driver: Star, flat, positive 9.7 Pressure gauge 9.8 Quick connect couplers/adapters 9.9 Pressure testing machine 9.10 Water pump
10. Leak testing method	10.1 Soap-sud solution 10.2 Using leak testing instrument 10.3 Observing pressure changes

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of plumbing fixtures and their functions 1.2 Procedure of installing toilet western or asian commode flange 1.3 Procedure of installing the western or asian commode and closet tank 1.4 Tightening techniques for western or asian commode to avoid damage/breakage 1.5 Care in the installation of ceramic type plumbing fixtures 1.6 Methods and procedures of installing different types of plumbing fixtures 1.7 Leak testing and qualifying workmanships in plumbing fixture installation 1.8 Understanding pressure testing methods, procedures, and tools, including allowable pressure levels and equipment accuracy. 1.9 Applying pressure progressively and conducting leak tests at various stages using appropriate methods. 1.10 Following safety precautions, reporting test results, and recording them as per workplace requirements. 1.11 Ensuring proper documentation and compliance with workplace pressure testing procedures.
2. Underpinning Skills	2.1 Gathering installed plumbing fixtures in the work site 2.2 Installing western or asian commode flange securely making sure the bolts are in place 2.3 Placing the new western or asian commode on the flange aligning the bolt holes with the bolts of the flange and wax ring in its proper place. 2.4 Tightening squarely the nuts with washers without overtightening 2.5 Installing the water closet tank on the western or asian commode in accordance with manufacturer's instruction 2.6 Installing other plumbing fixtures in accordance with plumbing plan and following manufacturer's

	instruction/specification 2.7 Checking newly installed plumbing fixtures for leaks and non-conformance with workplace requirements 2.8 Determining maximum test pressure and selecting appropriate pressure testing methods based on plumbing plans and workplace requirements. 2.9 Applying test pressures progressively and performing leak tests at each stage to identify leaks. 2.10 Reporting and recording test results while following safety precautions and workplace procedures. 2.11 Carrying out reworks /adjustment in accordance with workplace requirements
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Gathered installed plumbing fixtures in work site. 1.2 Installed western or asian commode flange securely making sure bolts are in place. 1.3 Placed the new western or asian commode on the flange aligning the bolt holes with the bolts of the flange and wax ring in its proper place. 1.4 Tightened squarely the nuts with washers without over tightening. 1.5 Installed the water closet tank on the bowl in accordance with manufacturer's instruction. 1.6 Installed other plumbing fixtures in accordance with plumbing plan and following manufacturer's instruction/specification. 1.7 Checked newly installed plumbing fixtures for leaks and non-conformance with workplace requirements. 1.8 Carried out reworks /adjustment in accordance with
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	workplace requirements.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT TROUBLESHOOTING IN PLUMBING SYSTEM	Nominal Duration: 26 Hrs.	Unit Code: SICIP-CON-PLU-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out troubleshooting in plumbing system. It specifically includes the tasks of identifying common plumbing system problems, preparing for troubleshooting work, troubleshooting plumbing system and cleaning/maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify common plumbing system problems	1.1 <u>Common plumbing system problems</u> are recognized. 1.2 Potential causes and consequences of each problem are explained. 1.3 Visual inspection is used to identify potential problems. 1.4 Pressure testing is used to identify leaks. 1.5 Pipes, fittings, and valves are checked for signs of wear, corrosion, or damage.
2. Prepare for troubleshooting work	2.1 Required <u>tools, equipment and materials</u> are identified for troubleshooting work. 2.2 Testing tools, equipment and materials are collected and checked for usability/accuracy. 2.3 <u>PPEs</u> are gathered and strictly used.
3. Troubleshoot plumbing system	3.1 Leaks are repaired using appropriate methods, such as, clamping or replacing. 3.2 Faulty components, including pipes, fittings, and valves, are replaced. 3.3 Pipes and fixtures are cleaned and flushed to remove clogs, debris and mineral buildup. 3.4 Water tank/ reservoir is cleaned using appropriate method.
4. Clean/maintain the work area	4.1 Workplace, tools and equipment are cleaned and checked for normal operation. 4.2 Waste materials are disposed in accordance to workplace requirement.

Range of Variables

Variable	Range (Includes but not limited to):
1. Common plumbing system problems	1.1 Leaks and water loss 1.2 Clogs and blockages 1.3 Low water pressure 1.4 Noise and vibration

	1.5 Corrosion and scale buildup 1.6 Issues with commode, commode accessories, bath tub, geyser
2. Tools and equipment	2.1 Measuring tape 2.2 Adjustable wrench 2.3 Box wrench set 2.4 Pipe wrench 2.5 Pliers, mechanical 2.6 Screw driver: Star, flat, positive 2.7 Pressure gauge 2.8 Quick connect couplers/adapters 2.9 Equipment 2.10 Pressure testing machine 2.11 Water pump 2.12 Pressure testing machine 2.13 Water pump
3. PPE	3.1 Hard hat (Skull guard) 3.2 Safety glass (receptacles) 3.3 Hand gloves 3.4 Safety shoes 3.5 Apron 3.6 Working clothes

Curricular Content Guide

1. Underpinning Knowledge	1.1 Common piping system problems (e.g., leaks, clogs, low water pressure, corrosion, etc.) 1.2 Causes and consequences of these plumbing issues 1.3 Visual and pressure testing methods to identify leaks and potential problems 1.4 Methods for repairing leaks, replacing faulty components, and cleaning pipes 1.5 Proper handling of water tanks and reservoirs for cleaning 1.6 Use of troubleshooting tools, equipment, and materials 1.7 Safety procedures and the proper use of PPE
2. Underpinning Skills	1.1 Identifying common plumbing system problems through visual inspection and pressure testing 1.2 Using appropriate tools to repair leaks and replace faulty components like pipes, valves, and fittings 1.3 Cleaning and flushing pipes and fixtures to remove clogs, debris, or mineral buildup 1.4 Effectively handling and cleaning water tanks/reservoirs 1.5 Using PPE and maintaining workplace safety 1.6 Cleaning and maintaining tools and equipment after troubleshooting work
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities

	3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Evidence that the candidate can identify plumbing system problems and explain their causes 1.2 Demonstrates the ability to prepare for and carry out troubleshooting tasks using the correct tools and materials 1.3 Effectively repairs leaks and replaces faulty components while adhering to safety and quality standards 1.4 Properly cleans and maintains the work area and equipment after completing the work
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Plumbing and Pipe Fitting
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops with internet facility	1
2.	Multimedia projector & screen	1
3.	Plumbing workstation/ booth with fitting fixture installation facility	5
4.	Kitchen Sink	2
5.	Water Basin	5
6.	Western commode	5
7.	Asian commode	5
8.	Urinal	2
9.	Shower unit	5
10.	Bath tub	1
11.	Geyser	2
12.	Hot and cold-water mixer tap	5
13.	Pipe vice (with stand/ work table)	16
14.	Pipe threading machine	1
15.	Thread cutter	16
16.	Angle grinder	10
17.	Pedestal grinder	2
18.	Hammer drill machine	5
19.	PPR welding machine	5
20.	Core cutting machine	1
21.	Plastic septic tank	1
22.	Water pump	5
23.	Overhead water Tank	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Plumbing and Pipe Fitting the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.